

MARTÍN RODRIGUEZ

Recent M.Sc. Graduate, Background in Embedded Systems, Signal Processing, and Machine Learning

📍 Portland, OR
✉ mtrpdx@gmail.com

🌐 mtrpdx.github.io

🌐 martintrodriguez

☎ +1 (503) 729-9373
📱 mtrpdx

EDUCATION

M.Sc. in Electrical and Computer Engineering, Portland State University 2024
Signal Processing and Machine Learning, GPA: 3.40

Thesis: Applying Positive Unlabeled Learning Techniques and Using the Kullback-Leibler Divergence to Improve Geothermal Surveying Assessments

B.Sc. in Electrical and Computer Engineering, Portland State University 2019
Embedded Systems, GPA: 3.13

TECHNICAL SKILLS

Programming: Python, NumPy, scikit-learn, PyTorch, TensorFlow, Matplotlib, librosa, OpenCV, MATLAB, C/C++, Julia, Swift, SwiftUI, bash, ARM/MIPS Assembly, $\text{\LaTeX}$

Software: LTSpice, Emacs, XCode, CrossCore Embedded Studio, Ableton Live, TouchDesigner, Git, Jira

General: Writing, Research, Signal Processing, Machine Learning, Statistical Analysis, Positive Unlabeled Learning, Bayesian Methods, Conformal Prediction, Embedded Systems, High Performance Computing, Sound Design, General Hacking

ENGINEERING/TECH EXPERIENCE

Quality Assurance (QA) Lead, Plus QA, Portland, OR Jan. 2021–Mar. 2022

- Worked with clients to develop comprehensive testing strategies and provide assistance to existing QA teams
- Led QA team for mobile marketing campaign that successfully reached millions of users

QA Tester, Plus QA, Portland, OR Jun. 2018–Dec. 2019

- Performed QA testing for mobile and web apps on a variety of platforms

Electrical Engineering Intern, Multiple Engineering Cooperative Program (MECOP) Mar. 2017–Dec. 2017
Lam Research, Tualatin, OR

- Studied methods of manufacturing and characterizing carbon probes for an atomic force microscope (AFM) via electron beam-induced deposition (EBID) with a focused ion beam scanning electron microscope (FIB-SEM), pointing to the possibility of improved tool sensitivity and efficiency
- Presented results at a company poster session
- Received the Lam Research Core Values Scholarship during my internship

Undergraduate Researcher, Research Experience for Undergraduates (REU) Jun. 2016–Sep. 2016
teuscher.:Lab, Portland, OR

- Optimized neural network (reservoir computation) techniques in Python and MATLAB, increasing accuracy and reducing simulation runtime
- Presented results at REU presentation session

Summer Intern, Oregon Space Grant Jun. 2011–Aug. 2011
NASA Goddard Space Flight Center, Greenbelt, MD

- Researched available MATLAB orbit simulation package capabilities, aiding in the nascent stages of the CubeSat (modular satellite systems for use in education) program

POSTER PRESENTATIONS

Rodriguez, M., Lipor, J.J., Mordensky, S.P., Burns, E.R., DeAngelo, J., 2024, *The Advantages of the Kullback-Leibler Divergence as an Evaluation Metric for Geothermal Favorability Prediction with Machine Learning*. Geothermal Rising Conference, Waikoloa, HI, 27–30 October 2024.

PROJECTS

Real-Time Tempo Detection with Harmonic-Percussive Source Separation via Median Filtering 2022
EE 522: Discrete Time Processing Final Project, Portland State University

Image Sorting and Sequencing using Canny Edge Detection and Hough Transforms 2022
EE 513: Intro to Image Processing Final Project, Portland State University

Early Detection of Forest Fires with Environmental Sensors, Computer Vision, and Deep Learning Techniques in Python and TensorFlow 2019
Capstone Project, Portland State University/Intel

ACHIEVEMENTS

Lam Research Core Values Scholarship Recipient 2017
Lam Research

Multiple Engineering Cooperative Program (MECOP) Intern 2017
Portland State University, Lam Research

Research Experience for Undergraduates (REU) Student 2016
Portland State University

Ronald E. McNair Scholar 2011
Portland State University McNair Scholars Program

Oregon Space Grant Recipient 2011
Oregon NASA Space Grant Consortium

SERVICE INDUSTRY EXPERIENCE

Pizza/Prep Cook, No Saint, Portland, OR Dec. 2023 – Current
• Prepare food before and during service including pizzas and desserts, maintain cleanliness of the work area

Line/Prep Cook (Salads and Desserts), Gumba, Portland, OR Jul. 2022 – Mar. 2024
• Prepared food before service and during service on the pantry station

Dishwasher/Prep Cook, Gado Gado, Portland, OR Aug. – Oct. 2019
• Washed dishes and performed light prep work

Dishwasher/Prep Cook, Clyde Common, Portland, OR Dec. 2017 – Jun. 2018
• Washed dishes and performed light prep work and maintenance for the restaurant

Host/Server/Bartender/Dishwasher, Ned Ludd, Portland, OR Oct. 2014 – Mar. 2016
• Began washing dishes, transitioned into performing host duties in the restaurant and serving and tending bar for the event space

Server/Bartender, El Sapo, Portland, OR Sep. 2013 – Aug. 2014
• Began serving in the restaurant, quickly transitioned to serving and bartending